



Project Title: Game Changer – Insights from Football Match and Player Data

➤ Business Context:

- You're working with a sports analytics startup that wants to unlock insights using historical football data. They aim to help coaches, clubs, and analysts in tactical decision-making, player scouting, and performance evaluation.
- Your job is to clean and transform raw data into actionable insights, define metrics, and explore match, player, and team dynamics.

➤ Files Provided:

- Match.csv
- Player.csv
- Player_Attributes.csv
- Team.csv
- Team_Attributes.csv
- Country.csv
- League.csv
- sql_sequence.csv (for maintaining primary key values)



Project Steps

➤ Data Exploration:

- Understand foreign key relationships
- Identify data types and missing values
- Explore target columns like home_team_goal, player_attributes, match_date, team attributes



➤ Data Cleaning Rules (Complex)

Components	Description
Normalize Dates	All dates to YYYY-MM-DD
Handle Nulls	Fill using forward-fill or remove based on relevance
Trim Strings	Remove extra spaces from names
Deduplicate Records	Based on composite keys (e.g. team_name + team_api_id)
Normalize Names	Convert league/team/player names to Title Case
Outlier Detection	Flag goal > 10 or < 0
Null Ratings	Fill missing player/team attributes using recent past entries
Validate Scores	Ensure goal columns contain only non-negative integers

➤ New Column Creation (Business Rules & Transformation Logic)

New Column	Source Table	Description
player_age	Player + Match	DATEDIFF (match_date, birthday) / 365.25
season	Match	Extracted from match_date as YYYY or range like 2011/12
goal_difference	Match	home_team_goal - away_team_goal
match_result	Match	Win, Draw, Loss based on goal_difference
team_strength_score	Team_Attributes	Sum of buildUpPlaySpeed + chanceCreationPassing + defencePressure
player_performance_index	Player_Attributes	Weighted avg: 0.3*dribbling + 0.3*ball_control + 0.2*short_passing + 0.2*vision
home_advantage_flag	Match	1 if home_team_goal > away_team_goal, else 0
formation_effectiveness	Match	Aggregate win % grouped by formation fields
clean_sheet_flag	Match	1 if opponent team scores 0, else 0
top_11_rank	Player_Attributes	Rank top 11 players by performance index in a team
team_form	Derived	Rolling 5-match result encoded as W/D/L for each team
league_goal_avg	Match + League	Group by league and season, avg(home + away goals)
star_player_flag	Player_Attributes	1 if rating > 85 and performance_index > threshold

New Column	Source Table	Description
is_undefeated	Match	True if team hasn't lost in last N games (based on form)
foul_intensity	Match	Fouls per match by team (if foul data exists)
experience_level	Player	Years between debut year and current season
pressure_rating	Team_Attributes	Label as 'Low', 'Med', 'High' based on defencePressure thresholds

➤ Business Questions (Use SQL Queries)

League & Team Performance:

- Top 3 teams with highest win rate per league/season?
- Avg goal per match per league?
- Which teams had the best defense (lowest avg goals conceded)?

Player Analytics:

- Who are the most consistent players (lowest stddev in rating)?
- What's the average age of top-performing players per league?
- Correlation between performance index and player height/weight?

Match Analysis:

- What is the average goal difference per team per season?
- Home vs Away advantage per league?
- Which team had the most comeback wins (trailing at halftime)?

Tactical Strategy:

- Impact of buildUpPlaySpeed on win %
- Do certain formations yield better results in specific leagues?
- Which teams changed formation styles and improved performance?

Advanced Analytics:

- Predict match result based on previous form, team attributes (SQL ML logic)
- Player clusters based on attributes (e.g., attackers, midfielders, defenders)
- Create a Player Retention Funnel — how long do players stay top-rated?

➤ Business Rules & Metric Definitions

Term	Definition
Win Rate	Matches won / Total matches
Clean Sheet	Match where opponent scores 0
Star Player	High performance index & rating
High Pressure Team	DefencePressure > 70
Build-Up Speed	Measured from buildUpPlaySpeed
Match Intensity	Based on fouls, cards, tackles
Performance Index	Weighted average of key skills
Team Form	Sequence of W-D-L over last 5 matches



➤ KPIs to Track

KPI	Description
Avg Goals per Match	Overall & League-wise
Win % per Team	Seasonal success
Avg Player Rating	For scouting
Home Advantage %	Home win rate
Player Index	Ranking system
Goal Conversion %	Goals / Shots
Clean Sheet Rate	% of matches with 0 goals conceded
/Formation Success %	Wins per formation



➤ Dashboard Suggestions (Tableau / Power BI)

Section	Visual
League Summary	Line chart of avg goals per season
Player Ratings	Heatmap of ratings by age & season
Team Form	Win/Loss/Draw trend lines
Formation Effectiveness	Bar chart or radar chart
Player Performance	Top 11 table with filters
Star Players	Scatterplot: Index vs Age



➤ Final Deliverables

Item	Description
Cleaned SQL DB	Structured & normalized
ER Diagram	Show relationships
SQL Scripts	For cleaning, transformation, KPIs
Insights Report	Executive summary
Dashboard	(Optional)
ML-ready Dataset	Final table with engineered features

